

ii) Silica :

A good brick earth should contain about 50 to 60% of silica. Silica exists in clay either as free or in combined form. As free sand, it is mechanically mixed with clay and in combined form, it exists in chemical composition with alumina. Presence of silica prevents cracking, shrinking and warping of raw bricks. It thus imparts uniform shape to the bricks. Durability of bricks depends on the proper proportion of silica in brick earth. Excess of silica destroys the cohesion between particles and brick will become brittle.



iii) Lime :

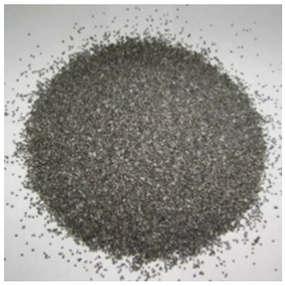
A small quantity of lime is desirable in finely powdered state to prevents shrinkage of raw bricks. Excess of lime causes the brick to melt and hence its shape is lost.

iv) Oxide of Iron :

A small quantity of oxide of iron to the extent of 5 to 6 % is desirable to impart red colour to bricks. Excess of iron oxide makes the bricks dark blue or blackish.

v) Magnesia :

A small quantity of magnesia in brick earth imparts yellow tint to bricks, and decrease shrinkage. But excess of magnesia leads to the decay of bricks.



The ingredients like iron pyrites, alkalies, pebbles, organic matter should not be present in good brick earth.

3.2.5 Manufacture of Brick

The following are various steps involved in the preparation of bricks :

1. Preparation of clay
2. Moulding
3. Drying
4. Burning

3.2.5.1. Preparation of Clay

The preparation of clay involves the following operations:

a) Unsoiling : Top layer of 20 cm depth is removed as it contain impurities.

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